How strong would the thing you cannot see have to be?

Students who take extra tutoring score higher, and the students who sign up were already ahead. When prior attainment is in the data this is a solved problem. When it is not — which is most school datasets — what is left to say?

Abstract

Students who take extra tutoring score higher, and the students who sign up were already ahead. When prior attainment is in the data this is a solved problem. When it is not — which is most school datasets — what is left to say? Case one: the confounder is recorded, so ask the graph what to condition on. 4 steps are recorded in full below. With the confounder measured, adjustment moved the estimate from 2.73 to 2.01 against a truth of 2.00. That is the easy case, and it is worth running first precisely because it calibrates the hard one: it shows what the missing column was worth. No assumptions were recorded for this run, which is not the same as none being required.

1. Introduction

This report addresses one question: Students who take extra tutoring score higher, and the students who sign up were already ahead. When prior attainment is in the data this is a solved problem. When it is not — which is most school datasets — what is left to say?

2. Methods

Question: Students who take extra tutoring score higher, and the students who sign up were already ahead. When prior attainment is in the data this is a solved problem. When it is not — which is most school datasets — what is left to say?

Case one: the confounder is recorded, so ask the graph what to condition on

An adjustment set is a property of the graph, not a matter of taste. The back-door criterion says exactly which variables block the non-causal paths, and minimal_adjustment_sets enumerates every sufficient set — so the choice of controls becomes a derivation rather than a habit.

Considered instead: Putting every available column on the right-hand side. That is the default in practice and it is unsafe in both directions: conditioning on a collider opens a path that was closed, and conditioning on a mediator removes part of the effect you were trying to measure.

```
graph : X -> Y, Z -> X, Z -> Y
status : identified via backdoor
condition on : ['Z']
minimal sets : [['Z']]
raw gap 2.726 [ 2.693, 2.758] error +0.726
back-door adjusted 2.013 [ 1.985, 2.040] error +0.013
```

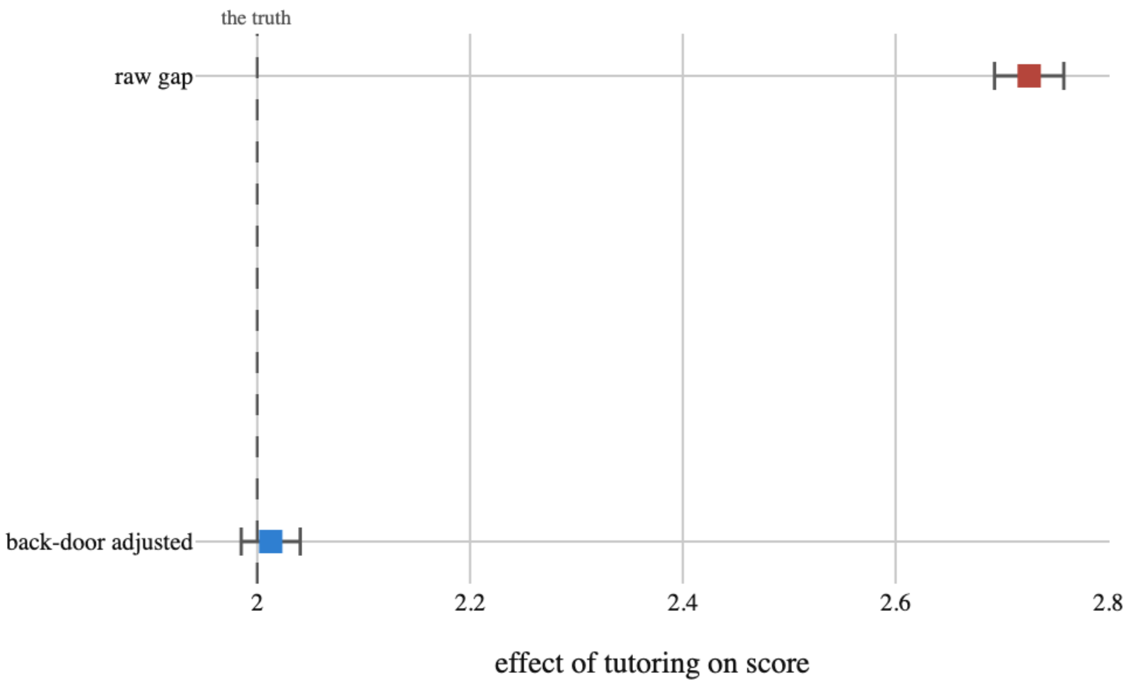

Figure 1. With prior attainment in the data, adjustment does the whole job. One column moves the answer from confidently wrong to correct. That is also the measure of what is lost in case two, where the column does not exist

Case two: it was never recorded, so the verdict is a refusal

The same graph with prior attainment unmeasured has no back-door adjustment set at all. axiom downgrades the verdict and names what is missing rather than returning a number with a caveat attached — a number with a caveat attached gets quoted; a refusal does not.

Considered instead: Fitting on the columns you have and putting 'unobserved confounding cannot be ruled out' in the limitations section. That sentence appears in almost every observational paper and changes no reader's behaviour, because it does not say how much confounding would matter.

```
columns you have : ['X', 'Y']
status : downgraded (backdoor_unmeasured)
would need : ['Z'], which you do not have
fitting anyway : 2.726 against a truth of 2.000 (+0.726)
```

Say how strong the invisible thing would have to be

An unmeasured confounder would have to explain 86.2% of the residual variance in both tutoring and scores to reduce the estimate to zero. Prior attainment — the very variable we hid — explains roughly 39% of tutoring and 56% of scores, which is nowhere near enough. So the honest reading is that this estimate survives a confounder as strong as the one we know about, and would only collapse under one substantially stronger. That is a claim a head of department can argue with; 'unobserved confounding cannot be ruled out' is not. The remaining honest question is not 'what is the effect' but 'how much confounding would it take to change the conclusion'. The robustness value is the share of residual variance in BOTH tutoring and scores that an unmeasured variable would have to explain to wipe the estimate out. It turns an unanswerable question into one a subject expert can actually judge.

Considered instead: Reporting the raw estimate with a wider interval to 'account for' confounding. Widening an interval models noise, and confounding is not noise: it moves the centre, and no amount of extra width puts it back.

```
robustness value : 86.2%
```

benchmark confounder	R2 with tutoring	R2 with score	estimate becomes
a tenth as strong as attainment	0.039	0.056	2.670
half as strong	0.195	0.281	2.419
as strong as attainment	0.390	0.562	2.022

Table 1

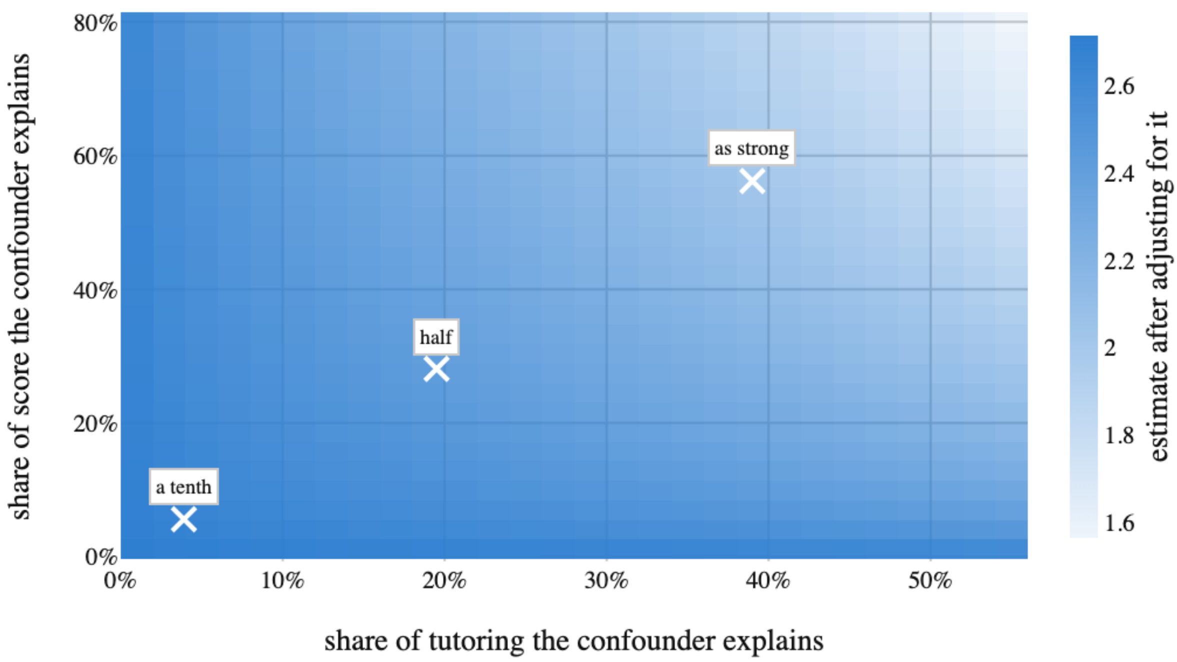

Figure 2. Every confounder that could exist, and what it would do to the answer. Darker is a larger surviving estimate. The three marks are confounders benchmarked against prior attainment itself, and even the strongest of them leaves the estimate at 2.02. The worst corner of this whole grid — a confounder explaining 55% of tutoring and 80% of scores — still leaves 1.56, comfortably above the 1 the decision turns on

Then ask the only question that pays: does any of it change the decision?

A finding is not a decision. The programme gets funded if the effect exceeds 1.0 with 90% certainty, so the useful sensitivity analysis walks the bias up until that call flips, rather than until the estimate reaches zero. Sensitivity should be measured against the threshold in play, not against the null.

Considered instead: Testing against zero. Nothing on this table was ever going to be decided at zero — a tutoring programme with an effect of 0.2 is as unfundable as one with an effect of 0, so a robustness value computed against zero answers a question nobody asked.

```
decision : fund if the effect exceeds 1.0
at zero bias : True (P = 1.000)
flips once bias : 1.8
real bias here : 0.726
```

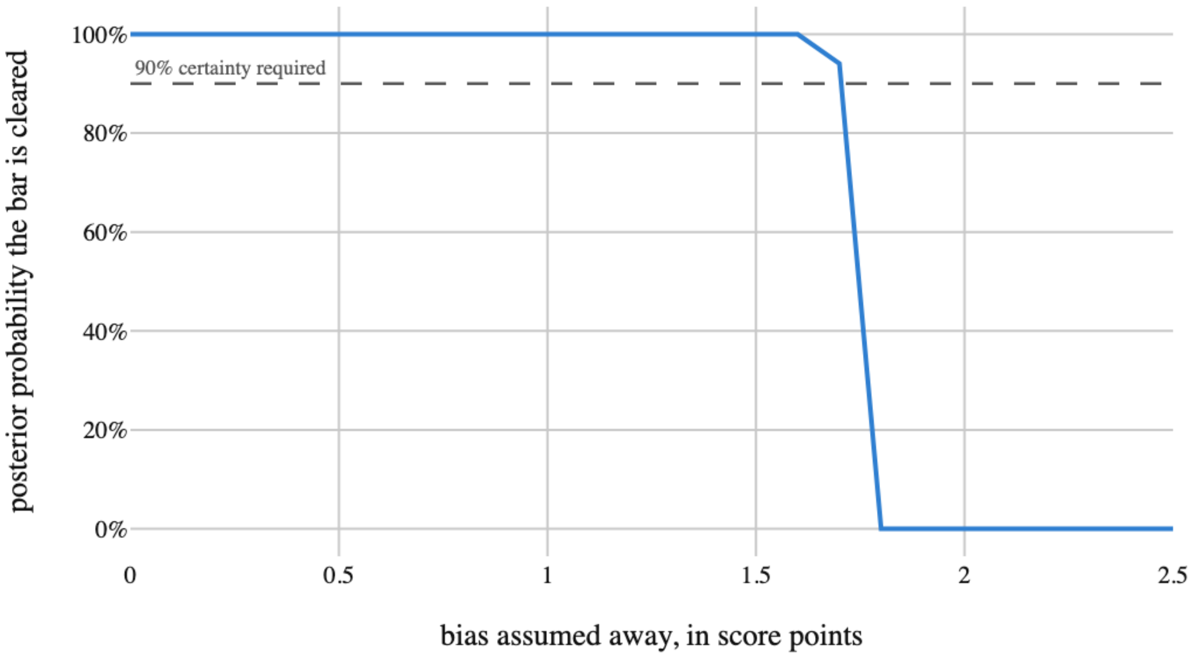

Figure 3. How much bias the funding decision can absorb. The call survives until an assumed bias of 1.8 score points. The bias actually present is 0.73 — a factor of 2.5 of headroom, which is a far more useful sentence than either 'it is significant' or 'we cannot know'

3. Results

No findings were recorded.

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

With the confounder measured, adjustment moved the estimate from 2.73 to 2.01 against a truth of 2.00. That is the easy case, and it is worth running first precisely because it calibrates the hard one: it shows what the missing column was worth.

With it hidden, no estimator recovers the truth — the estimate sits +0.73 out and stays there. What replaced the point estimate was not a wider interval but a different kind of statement: a confounder would need to explain 86.2% of both sides to erase the effect, and 1.8 points of bias to change the funding call.

This one survives. That is not the interesting part — the interesting part is that it survives by an amount anyone can check. Two different sensitivity questions, one against zero and one against the funding bar, both came back with room to spare, and both came back as quantities rather than as a hedge in a limitations paragraph.

It would not always. Run the same three steps on a weaker programme and the contour's marks move into the pale region, the tipping bias falls below the bias a benchmark confounder would produce, and the correct output is 'do not fund on this evidence'. The value of the machinery is that it returns that answer as readily as this one.

5. Model checking

No diagnostics were recorded. An analysis with no model checking is not therefore sound; it is unexamined.

6. Discussion

There are no findings to interpret.

7. Conclusions

Students who take extra tutoring score higher, and the students who sign up were already ahead. When prior attainment is in the data this is a solved problem. When it is not — which is most school datasets — what is left to say?

No finding was recorded, so no conclusion follows.

8. Limitations

No assumption in the record is left unresolved. That is a statement about the record, not a guarantee about the world.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Field	Value
field	Education
source	examples/ walkthrough record
steps	4
evidence hash	2b9928daf7d6bc916f829c79ae90b20e10cdfcf0304a1dfc61a9a7561ee8322d

Table 2. Run